PORTFOLIO ANALYTICS | 2026

STOCK MARKET BUSINESS STRATEGY

HDFC · INFOSYS · RELIANCE | 12-Month Analysis

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Data Analyst · Power BI · SQL · Python

233M	1.00K	3
Total Volume	Avg Price	Stocks

AGENDA

What we cover today

Today's Presentation

01 **Project Overview**
Objectives, dataset & tech stack

02 **Data Pipeline**
Python → SQL → Power BI workflow

03 **Stock Performance**
HDFC, Infosys & Reliance comparison

04 **Key Insights**
Volatility, returns & trading peaks

05 **Business Strategy**
Investment & trading recommendations

06 **Future Roadmap**
Enhancements & next steps

PROJECT OVERVIEW

End-to-End Stock Market Analytics

Objectives

- Analyze HDFC, Infosys & Reliance
- 12-month return & risk assessment
- Identify best long-term stock
- Find peak trading windows
- Executive-level dashboard

Dataset

- 3 major Indian stocks
- 12 months of daily data
- Fields: Open, High, Low, Close
- Volume & Daily Return
- Total Volume: 233M units

Tech Stack

- Python — Data cleaning
- SQL Server — 10+ queries
- Power BI — Dashboard
- DAX — KPI measures
- Jupyter Notebook

DATA ANALYTICS PIPELINE

Python → SQL → Power BI

01 Python
Data Cleaning

- Import raw CSV data
- Handle nulls & duplicates
- Engineer daily_return
- Format date columns
- Export clean dataset

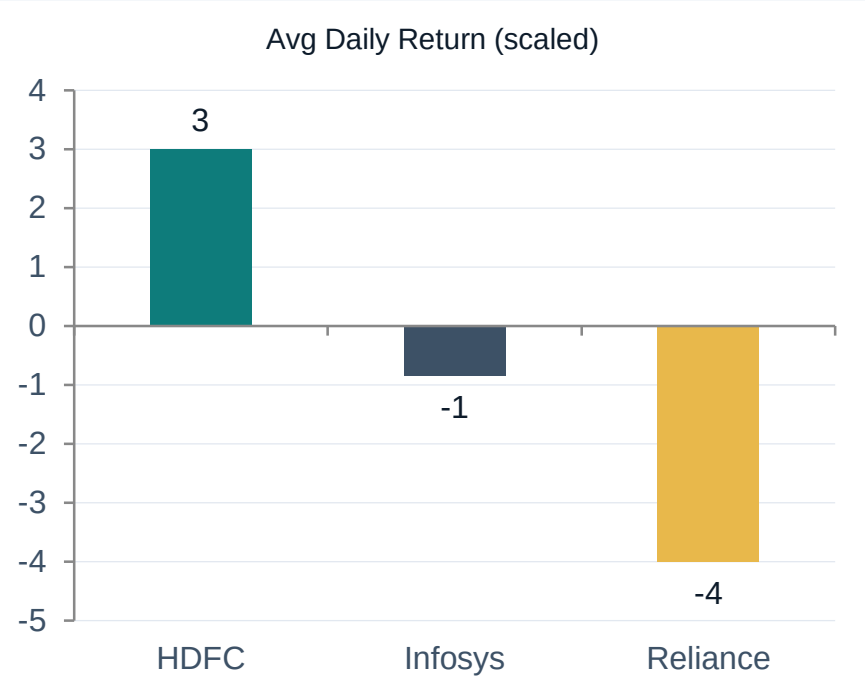
02 SQL Server
Analysis (10 Queries)

- Avg return per stock
- Volatility via STDEV()
- Top 5 worst trade days
- Seasonal month trends
- CTE: total return %

03 Power BI
Dashboard & DAX

- 6 KPI metric cards
- Bar, donut & line charts
- DAX calculated measures
- Dynamic filters
- Executive insight card

STOCK PERFORMANCE COMPARISON



HDFC

Verdict: **Best Pick**

Return	Volatility	Contribution
+0.0003	**0.011**	**100%**

Infosys

Verdict: **Neutral**

Return	Volatility	Contribution
-0.000085	**0.011**	**Minimal**

Reliance

Verdict: **Avoid**

Return	Volatility	Contribution
-0.0004	**0.010**	**Negative**

KEY INSIGHTS

What the data tells us about these 3 stocks

HDFC is the top performer

Only stock with positive avg daily return (+0.0003). Contributes 100% of portfolio returns — the clear long-term hold.

Reliance drags portfolio negative

Avg daily return of -0.0004 pushes overall portfolio to -0.0051%. Worst performer across all 3 metrics.

May–June is the peak trading window

All 3 stocks spike to ~8M volume in months 5–6 vs a ~6M baseline. Best period for high-liquidity trade execution.

Volatility is macro-driven

HDFC & Infosys share identical volatility (0.011) and Reliance is 0.010 — showing uniform market-wide pressure, not stock-specific risk.

BUSINESS STRATEGY & RECOMMENDATIONS

Long-Term Investment

Recommended: HDFC

- Allocate majority of long-term portfolio to HDFC
- Only stock with consistent positive returns
- 100% return contribution — proven performer
- Volatility is manageable at 0.011

Trading Strategy

Timing: May–June Window

- Execute high-volume trades in May–June
- ~8M volume spike across all 3 stocks
- Tighter bid-ask spreads during peak volume
- Ideal for both entry and exit positions

Portfolio Management

Risk-Adjusted Allocation

- Overweight HDFC (top performer)
- Hold Infosys as low-risk neutral position
- Reduce Reliance exposure — negative return drag
- Max drawdown of -0.03 — set stop-loss at this level

SQL ANALYSIS — 10 BUSINESS QUESTIONS

#	Business Question	SQL Technique
01	Average daily return per stock	AVG(daily_return) GROUP BY stock_name
02	Most volatile stock	STDEV(daily_return) GROUP BY stock_name
03	Total return per stock	SUM(daily_return) ORDER BY desc
04	Highest average return month	TOP 1 + ORDER BY highest_avg_return
05	Yearly average price trend	AVG(close) GROUP BY year
06	Stock with highest trading volume	MAX(volume) GROUP BY stock_name
07	Top 5 worst return days	TOP 5 ORDER BY daily_return ASC
08	Stock performance comparison	AVG + STDEV GROUP BY stock_name
09	Seasonal return trends by month	SUM(daily_return) GROUP BY month
10	Best stock for long-term (CTE)	CTE: Initial vs Current Price → Total Return %

DASHBOARD KPIs

Power BI — Key Performance Indicators

Avg Closing Price

1.00K

Average across 12 months

Avg Daily Return

-0.0051%

Portfolio daily avg

Total Volume

233M

Units traded across all stocks

Max Drawdown

-0.03

Worst peak-to-trough decline

Best Stock

HDFC

100% return contribution

Worst Stock

Reliance

Negative return drag

FUTURE ROADMAP

Phase 1
Live Data Integration

- Connect Yahoo Finance API
- Real-time price streaming
- Auto-refresh Power BI dashboard

Phase 2
Predictive Analytics

- LSTM price forecasting model
- Linear regression for returns
- Confidence interval bands

Phase 3
Portfolio Expansion

- Expand to 10+ stocks
- Sector-wise comparison
- NIFTY 50 benchmark overlay

Phase 4
Stakeholder Reporting

- Automated monthly PDF reports
- Email alerts on threshold breach
- Executive summary email digest

THANK YOU

Questions & Discussion

Prateek Bag · Data Analyst

Python · SQL · Power BI · DAX

3	12	10+	233M
Stocks	Months	SQL Queries	Volume